Data types

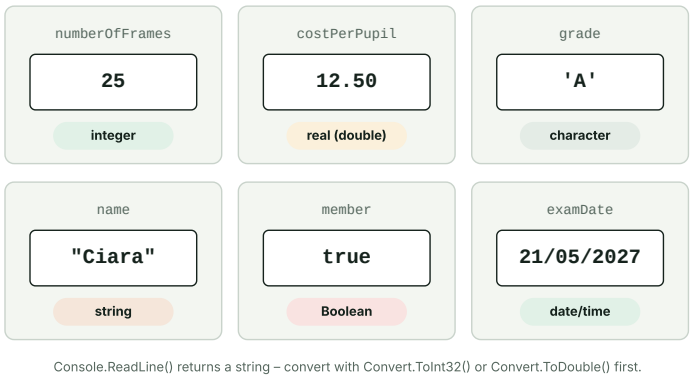
D07
FACT FILE

Specification: students should be able to

- Describe and use appropriately the following data types: numeric (integer and real), date/time, character and string

CCEA fact file: Digital Data 2

Key words: data type, integer, real, double, character, string, Boolean, date/time, casting, type conversion



Six variables, each holding a value of a different data type

Data types

An integer stores a whole number; a real stores a number with a decimal part (double in C#); a character stores one symbol; a string stores text; a Boolean stores one of two values, true or false; date/time stores dates and times. The type controls how much memory is used and which operations make sense. In C#, Console.ReadLine() returns a string, so values must be converted with Convert.ToInt32() or Convert.ToDouble() before arithmetic. Dividing one int by another gives a whole number, so cast to double when a decimal answer is needed.

T Key terms from the mark schemes

Boolean
Data type with only two values, True or False.
Credited: U4 2024 Q8, 2025 Q6

Float / real
Data type for decimal numbers.
Credited: U4 2024 Q3

C C# example

Type conversion (casting)

```
Console.Write("Age? ");
int age = Convert.ToInt32(Console.ReadLine());
Console.WriteLine("Next year you will be " + (age + 1));

double price = Convert.ToDouble("3.99");
Console.WriteLine(price * 2);

int marks = 7;
double average = (double)marks / 2;   // cast: 3.5, not 3
Console.WriteLine(average);
```

Exam focus

Every paper asks for data types in context: 2019 Q1(a) and Q9(a)(ii), 2022 Q2(a) and Q4(b)(ii), 2023 Q3(a), 2024 Q3(b)(iii) and Q8(a), 2025 Q1(a) and Q6(b). Justify by linking to the data, e.g. 'real because the cost includes pence'.

Unit 4 2025 Q6	[25]	Tent hire algorithm; Boolean; constants; validation true/false; Y/N validation loop
Unit 4 2025 Q3	[15]	Musician hire algorithm; variables and data types; IDE aids; syntax vs logic
Unit 4 2025 Q1	[5]	String type, compound AND, IF output, NOT/AND table, bubble sort
Unit 4 2024 Q8	[18]	String and Boolean types; unit testing; test plan; black vs white box
Unit 4 2024 Q3	[17]	Travel insurance cost algorithm; validation; real data type; WHILE range-check loop

? Check yourself

1. State the most suitable data type for: (a) number of nights booked (b) cost of a booking (c) whether equipment is hired (d) customer name. [4]
2. Describe the Boolean data type. [1]
3. Explain why a race time of 12.84 seconds should be stored as a real. [1]

Answers: 1. (a) integer (1) / (b) real (accept float/double/currency) (1) / (c) Boolean (1) / (d) string (1) 2. A Boolean can hold only one of two values: True or False (1) 3. The time has a decimal (fractional) part, which an integer cannot store, so a real is needed (1)